

Total No. of Questions : 6]

SEAT No. :

P3701

[Total No. of Pages : 2

Engg. - 48

T.E. (Computer Engineering) (Semester - I)

**DATA COMMUNICATION AND WIRELESS SENSOR NETWORK (In Sem.)
(2012 Pattern)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) *Answer any three questions.*
- 2) *Solve Q-1 or Q-2, Q-3 or Q-4, Q-5 or Q-6.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume Suitable data if necessary.*

- Q1)** a) What is block coding explain any one method in detail. [5]
b) Explain wireless router, Access points, switches and hubs. [2]
c) What is VPN? What are applications of VPN? [3]

OR

- Q2)** a) Represent 1100100111 using following line coding schemes. [6]
i) AMI
ii) Manchester code
iii) Polar NRZ
b) With the help of waveform explain sampling, quantization and encoding. [4]

- Q3)** a) “In Go-Back-N ARQ, the size of the send window must be less than 2^m ”, with the help of flow diagram justify the statement? [5]
b) What is FHSS? Explain with diagram. [5]

OR

- Q4)** a) Draw an explain block diagram of Direct Sequence Spread Spectrum. [5]
b) Compare packet switching and circuit switching. [5]

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- Q5)** a) Draw and explain components of wireless sensor network. [5]
b) Draw block diagram for RFID reader and RFID tag. [3]
c) What are the advantages of RFID over bar code? [2]

OR

- Q6)** a) Write and explain characteristics of wireless sensor network. [4]
b) Define sensor network. Draw basic architecture of it and list its applications. [4]
c) What is an importance of sensors in robots? [2]

